A Kronecker-Set Unitary Is Weakly Supercyclic but Not Weakly Stable

Literature-implied answer to arXiv:2010.13331

13 August 2026

Status. Literature-implied full negative answer to the source's broad question. The supporting authors did not identify their construction as an answer to this later question.

Question and identification

Kubrusly–Vieira ask on page 5 whether every weakly supercyclic power-bounded operator is weakly stable.

> Lemma 3.1 is naturally extended to normed spaces by replacing inner product with dual pairs, and Hilbert-space adjoints with normed-space adjoints.
>
> 4. Weak Supercyclicity and Weak stability
>
> It was shown in [1, Theorem 2.2] that *if a power bounded operator on a Banach space is supercyclic, then it is strongly stable*. Such a result naturally prompts the question: *does weak supercyclicity imply weak stability for power bounded operators*? The question was posed and investigated in [9], and remains unanswered even if Banach-space power bounded operators are restricted to Hilbert-space contractions, where the problem is equivalently stated for unitary operators (see Remark 4.1 below), and also if weak supercyclicity is strengthened to weak l-sequential supercyclicity: *does there exist a weakly unstable and weakly l-sequentially supercyclic unitary operator*?

Figure 1: The question in arXiv:2010.13331, page 5.

Bayart–Matheron Example 3.6 supplies the counterexample. Let $E \subset \mathbb{T}$ be a perfect Kronecker set, let μ be a continuous probability measure supported on E, and let

$$U = M_z : L^2(\mu) \to L^2(\mu), \qquad (Uf)(z) = zf(z).$$

They prove that $\{az^n : a \in \mathbb{C}, n \geq 0\}$ is weakly dense in $L^2(\mu)$, so 1 is a weakly supercyclic vector for the unitary U.

The same displayed construction says that a Kronecker set is a Dirichlet set: there are positive integers $p_k \to \infty$ such that

$$\sup_{z \in E} |z^{p_k} - 1| \longrightarrow 0.$$

Consequently

$$\langle U^{p_k} 1, 1 \rangle = \int_E z^{p_k} \, d\mu(z) \longrightarrow 1,$$

so $U^n 1$ does not converge weakly to 0. Thus U is unitary (hence power bounded), weakly supercyclic, and not weakly stable.

Scope and search evidence

This implication answers the broad weak-density question negatively. It does not make the example weakly l-sequentially supercyclic: a weakly convergent sequence from a fixed projective unitary orbit has bounded

> We postpone the proof of the lemma, and now give the promised example. Recall that a compact set $E \subset \mathbb{T}$ is said to be a *Kronecker set* if the characters of $\mathbb{T}$ are uniformly dense in $\mathcal{C}(E, \mathbb{T})$, the set of all continuous functions on E with constant modulus 1. For further details on Kronecker sets and other thin sets from harmonic analysis, we refer the reader to [**10**]. It is well known, and easy to prove by Baire category arguments, that there exist perfect Kronecker sets. Let E be such a set, and let μ be any continuous Borel probability measure with support E. Finally, let $T = M_z$ be the multiplication operator acting on $L^2(\mu)$. We show that the set $\mathcal{D} = \{\lambda z^n;\ \lambda \in \mathbb{C},\ n \in \mathbb{N}\}$ is weakly dense in $L^2(\mu)$, which means that the constant function $\mathbf{1}$ is a weak supercyclic vector for T. Since E is a Kronecker set, it is also a *Dirichlet set*, which means that the constant function $\mathbf{1}$ a uniform limit (on E) of a sequence z^{p_k}, with $p_k \in \mathbb{N}$ and $p_k \to \infty$. Indeed, since

Figure 2: Bayart–Matheron, page 9: weak density and the Dirichlet subsequence appear in the same construction.

scalar multipliers, which is a materially stronger restriction than weak density. The companion full-result packet proves that no unitary of dimension greater than one has that direct-sequence property.

A bounded search through 13 August 2026 checked the run indexes, exact wording, the 2024 survey arXiv:2405.02752, arXiv:2306.08197, and the cited Bayart–Matheron and Shkarin papers. Later sources continue to call the broad question open; no source found explicitly records the two-line implication above. This packet therefore records an agent-identified literature implication, not a claim of a new construction.

Human-review recommendation. Check only the sign convention in the displayed inner product and confirm that the positive Dirichlet sequence p_k in Bayart–Matheron is the same sequence shown in the supporting excerpt. Either convention gives convergence to 1 after complex conjugation.

References

[1] C. S. Kubrusly and P. C. M. Vieira, *Boundedly Spaced Subsequences and Weak Dynamics*, Journal of Function Spaces (2018), Article ID 4732836; arXiv:2010.13331.

[2] F. Bayart and E. Matheron, *Hyponormal operators, weighted shifts and weak forms of supercyclicity*, Proc. Edinburgh Math. Soc. 49 (2006), 1–15, doi:10.1017/S0013091504000975.